



What's the Difference between Deep Learning, Machine Learning and AI?

AI, deep learning and machine learning are often mistaken for each other. Take a look at how they differ in this interesting article.

From being dismissed as science fiction to becoming an integral part of multiple, wildly popular movie series, especially the one starring Arnold Schwarzenegger, artificial intelligence has been a part of our life for longer than we realise. The idea of machines that can think has widely been attributed to a British mathematician and WWII code-breaker, Alan Turing. In fact, the Turing Test, often used for benchmarking the 'intelligence' in artificial intelligence, is an interesting process in which AI has to convince a human, through a conversation, that it is not a robot. There have been a number of other tests developed to verify how evolved AI is, including Goertzel's Coffee Test and Nilsson's Employment Test that compare a robot's performance in different human tasks.

As a field, AI has probably seen the most ups and downs over the past 50 years. On the one hand it is hailed as the frontier of the next technological revolution, while on the other, it is viewed with fear, since it is believed to have the potential to surpass human intelligence and hence achieve world domination! However, most scientists agree that we are in the nascent stages of developing AI that is capable of such feats, and research continues unfettered by the fears.

Applications of AI

Back in the early days, the goal of researchers was to construct complex machines capable of exhibiting some semblance of human intelligence, a concept we now term 'general intelligence'. While it has been a popular concept in movies and in science fiction, we are a long way from developing it for real.

Specialised applications of AI, however, allow us to use image classification and facial recognition as well as smart personal assistants such as Siri and Alexa. These usually leverage multiple algorithms to provide this functionality to the end user, but may broadly be classified as AI.

Machine learning (ML)

Machine learning is a subset of practices commonly aggregated under AI techniques. The term was originally used to describe the process of leveraging algorithms to parse data, build models that could learn from it, and ultimately make predictions using these learnt parameters. It encompassed various strategies including decision trees, clustering, regression, and Bayesian approaches that didn't quite achieve the ultimate goal of 'general intelligence'.